

FRONTEND SYSTEM DESIGN IN 30 DAYS →

DAY 18: REACT FIBER, RECONCILIATION & VIRTUAL DOM



Sibananda



THE REAL PROBLEM

? Why does changing a small input cause a full re-render?

✗ Inefficient updates → lags, performance drops

✓ Understanding Fiber helps you build snappy UIs

Let's decode how React's rendering engine works 💡



Sibananda



REACT'S VIRTUAL DOM (V-DOM)

- ✓ React creates a lightweight in-memory copy of the DOM
 - ✓ On every state/prop change → it diffs new V-DOM vs old V-DOM
 - ✓ Only the changed nodes are updated in the real DOM
- ⚙ Efficient → but large apps need something faster...



Sibananda



WHAT IS REACT FIBER?

- 🔍 React Fiber is the engine behind React 16+
- ✅ Breaks rendering into chunks → async & interruptible
- ✅ Improves responsiveness (esp. in large apps)
- ✅ Supports prioritization (urgent vs low-priority updates)
- 🧵 Think of it as: rendering in smaller units → better control



Sibananda



RECONCILIATION EXPLAINED

🧠 Reconciliation = how React updates the UI efficiently

✓ Compares current Fiber tree to the new one

✓ Decides which nodes to update, reuse, delete, or create

👁️ Optimization tip: Use keys in lists to help React track nodes!



Sibananda



KEY BENEFITS OF FIBER ARCHITECTURE

- ✓ Asynchronous rendering
 - ✓ Pausable & resumable rendering
 - ✓ Better performance for animations & transitions
 - ✓ Smarter scheduling = smoother UX
- 🚀 Helps React apps scale without UI jank



Sibananda



WHEN THIS KNOWLEDGE HELPS

- ✓ Debugging performance issues
- ✓ Choosing where to memoize / split components
- ✓ Understanding why React re-renders what it does
- ✓ Building UI that feels instant even with heavy logic



Sibananda



SUMMARY + TEASER

📌 Virtual DOM = fast diffing

📌 Fiber = async rendering engine

📌 Reconciliation = smart diffing & scheduling

📌 Use keys + memo + proper state handling for max benefit

👉 Up next in Day 19: Bundle Optimization – Webpack & Vite Best Practices

💬 Ever optimized your React app based on reconciliation?
Let's discuss 🙌



Sibananda